Position: World Models Should Be Benchmarked for Action–State Consistency

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Abstract

World models are increasingly used as simulation backbones for reasoning, planning, and interaction. Yet most evaluations still prioritize perceptual fidelity under passive rollouts, leaving action-conditioned dynamics under-specified. We argue that world models should be benchmarked for action–state consistency: whether interventions induce reliable, causally legible state transitions. We contribute a state-modality x action-interface taxonomy, a metric quad that separates state consistency and diversity from action rationality and branching control, and an explicit action-graph construction (AC-World) that makes these principles operational. We also define a minimal, code-backed diagnostic that measures action inference on a small branching world, illustrating how action-centric evaluation remains well-posed when prompt-only rollouts do not.

1. Introduction

World models aim to capture *how the world changes*: not only what is observed, but how latent state evolves over time and how interventions shape future outcomes. This ambition is no longer confined to model-based reinforcement learning (MBRL)—it now spans text-to-video generation (Sora, Veo) (OpenAI, 2024b; Google DeepMind, 2024), history-conditioned video simulation and interactive world models (PAN, Genie, Oasis, GameNGen) (Team, 2025a; Bruce et al., 2024b; Decart AI, 2024b; Valevski et al., 2024b), 3D and spatial world construction (World Labs Marble) (Li et al., 2024), and robotics or driving systems that couple perception to action (RT-2, UniSim, GAIA, Cosmos) (Brohan et al., 2023b; Yang et al., 2024b; Hu et al., 2023b; NVIDIA Research, 2025a). The community increasingly talks about these systems in a shared vocabulary of "world models," often implying a common scale of progress.

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The problem is that *progress is currently easy to claim but hard to compare*. "World model" describes incompatible objects: latent dynamics used for planning in Dreamer-style agents, pixel-level simulators for games, and prompt-only video generators with no stable action interface. Without a shared notion of state and action, papers talk past each other, and benchmarks reward the wrong properties. This mismatch is not cosmetic. Agents do not merely *watch* the world; they *intervene*. A system can produce visually coherent videos yet fail as a world model if interventions do not induce reliable, context-appropriate changes.

We argue that the field is missing two foundations. First, there is no widely adopted taxonomy that organizes world models by what they represent as *state* (text, images, video, learned latents, or explicit spatial structure) and how they accept or expose *actions* (none/prompt-only, pixel-level control, continuous motor commands, discrete symbolic actions, or hybrid mixes). This absence produces apples-to-oranges comparisons: a high-fidelity generator may look impressive yet be unusable for interaction; a controllable simulator may enable planning yet appear visually crude. Second, there is no principled evaluation axis that directly tests what makes a model a *world* model rather than a passive predictor: whether it preserves coherent, *action-conditioned* transitions under interventions.

Our position is simple: **world models should be evaluated by action–state consistency, not only by perceptual fidelity under passive rollouts.** This does not mean visual realism is irrelevant; it means realism is an incomplete proxy for the properties required by planning, tool use, and safety verification. We therefore advocate an action-centric framing that separates (i) how well the world holds together and how rich it is, from (ii) whether actions are causally legible and whether they provide meaningful branching control.

As a position paper, we make three contributions:

- **Taxonomy (Section 2).** We first define the core components of a world model (state, action, transition, agent) as a shared discussion ground, and show how this yields a practical state-modality $\times$ action-interface taxonomy (Ke et al., 2023).
- **Missing axis (Section 3).** We argue that action–state consistency is missing from dominant benchmarks, and

propose an action-centric evaluation lens (metric quad) that includes inverse-action rationality and controllable branching (Zhang et al., 2018; Unterthiner et al., 2019; Du et al., 2023).

- **Existence proof (Section 4).** We describe an explicit action-graph construction (AC-World) that makes states, actions, and transitions groundable into text, images, and video for concrete diagnostics.

We include one minimal diagnostic (Section 5) that our current implementation can run end-to-end, illustrating that action-centric evaluation is measurable today.

1.1. Definitions and Notation

We model an environment as a controlled dynamical system with *state space* $\mathcal{S}$ and *action space* $\mathcal{A}$. At time t, the environment is represented by a state $s_t \in \mathcal{S}$. An intervention $a_t \in \mathcal{A}$ induces a (possibly stochastic) transition

$$\mathcal{T} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S}), \qquad s_{t+1} \sim \mathcal{T}(s_t, a_t),$$

where $\Delta(\mathcal{S})$ denotes a distribution over next states. A benchmark instance is $\mathcal{I} = (s_0, G, \mathcal{C}, T)$ with initial state s_0, horizon T, a goal predicate $G : \mathcal{S} \to \{0, 1\}$, and optional dynamic constraints $\mathcal{C}$ enforced along a rollout. A world-model agent consists of a learned *world model* M_θ and, optionally, a *policy* π_ϕ:

$$M_\theta : (h_t, a_t) \mapsto p_\theta(s_{t+1} \mid h_t, a_t), \qquad a_t \sim \pi_\phi(\cdot \mid h_t),$$

Throughout, we use "state" to mean the step-to-step representation exposed to an evaluator (text, image, video, latent, or spatial/3D); the taxonomy below makes this representation choice explicit.

2. A Taxonomy for World Models

2.1. World model components: a unified discussion ground

The term "world model" is used across communities to describe systems with different interfaces and different objects of prediction. To keep comparisons precise, we refer to the notation in Section 1.1: an environment evolves as $s_{t+1} \sim \mathcal{T}(s_t, a_t)$, and a world model M_θ is any learned predictor that can be rolled out under interventions a_t (either because it directly models $p_\theta(s_{t+1} \mid h_t, a_t)$, or because it implements an equivalent simulator). A policy π_ϕ may be explicit (selecting $a_t \in \mathcal{A}$) or implicit (e.g., prompt-based control), but in all cases it induces trajectories by repeated action selection and state update.

This decomposition is intentionally generic; its value is that it makes two questions unavoidable for benchmarking: *what counts as state* (i.e., what representation is carried forward as s_t), and *what counts as action* (i.e., what interface defines admissible interventions in $\mathcal{A}$). These choices determine what a "rollout" means, what errors are diagnosable, and what metrics are well-posed.

2.2. Taxonomy axes derived from the components

With the above decomposition, two evaluation-critical axes become explicit. *State modality* is the type of representation used for s_t (text/image/video/latent/spatial), and *action interface* is how interventions a_t are specified and constrained (none/prompt-only, pixel, motor/continuous, symbolic, hybrid). The taxonomy is intentionally simple: it does not attempt to capture every modeling choice (e.g., diffusion vs. autoregressive), but it does capture the two degrees of freedom that most directly determine *what can be evaluated*. The same system can be strong under one axis and weak under another; without making both axes explicit, evaluation confounds representation choices with interface choices.

2.3. State modality: how is state exposed?

State modality specifies the representational layer at which the world is exposed to the evaluator: *text* (language state descriptions), *image* (single-frame visual state), *video* (temporal visual state), *latent* (learned compact state sufficient for prediction/control), and *spatial/3D* (explicit geometry or persistent spatial structure). This axis matters because state is not merely a snapshot: it encodes what is assumed to be tracked and preserved across time. In MBRL, latent states are learned to be predictive under actions (PlaNet, Dreamer) (Hafner et al., 2019; 2020; 2021), while in modern generative models, the state may be an image or video stream optimized for perceptual realism (OpenAI, 2024b; Google DeepMind, 2024; Wu et al., 2023; Lu et al., 2024; Tan et al., 2024). In spatial world modeling, state may include an explicit, editable representation of a scene with persistent structure (Li et al., 2024).

This axis also clarifies the role of representation learning: methods such as DINO, CLIP, and JEPA-style predictors (Caron et al., 2021; Radford et al., 2021; LeCun, 2022; Assran et al., 2023; Bardes et al., 2024) provide *state encoders* and predictive features that can support world modeling, but by themselves do not define an interactive world model without an action interface and a transition semantics.

2.4. Action interface: what counts as an "action"?

Action interface specifies how interventions are represented and interpreted by the system: **none** (prompt-only or passive generation), **pixel** (low-level control like mouse/keyboard or pixel deltas), **motor** (continuous control signals), **symbolic** (discrete structured commands), and **hybrid** (structured high-level actions with optional low-level condition-

ing). This axis matters because it determines whether the system can be used as an environment for planning: actions must have stable semantics across time and contexts.

Classical control and robotics benchmarks naturally inhabit motor/symbolic interfaces (Fu et al., 2021; Yu et al., 2020; James et al., 2020; Li et al., 2022). Recent interactive world models explore pixel-level control (Bruce et al., 2024b; Decart AI, 2024b; Valevski et al., 2024b), while others use natural language as an action channel (Team, 2025a; Yang et al., 2024b). Prompt-only video generators occupy the none/prompt-only interface: they may accept instructions, but they do not expose a stable, stepwise action semantics that would allow comparable evaluation across multi-step rollouts.

2.5. Why these axes matter: comparability and failure attribution

The consequence of the taxonomy is methodological: evaluation should be reported *within each cell* (m_S, m_A), rather than across cells. Within a cell, we can define consistent inputs/outputs, comparable action sets, and meaningful metrics. Across cells, differences can be attributed to interface regimes rather than to the quality of a model.

Table 1 illustrates representative systems. Prompt-only video generators such as Sora and Veo (OpenAI, 2024b; Google DeepMind, 2024) reside in the video/none cell, while interactive models such as Genie, Oasis, and GameNGen (Bruce et al., 2024b; Decart AI, 2024b; Valevski et al., 2024b) occupy video/pixel or video/symbolic interfaces. PAN uses natural language as a structured action channel for video simulation (Team, 2025a). Robotics and autonomous driving systems such as RT-2, UniSim, and GAIA couple perception with motor or hybrid actions (Brohan et al., 2023b; Yang et al., 2024b; Hu et al., 2023b). This taxonomy makes explicit why "visual realism" and "action controllability" are often confounded: many systems optimize one axis while leaving the other under-specified.

3. Why Action–State Consistency Is the Missing Axis

Current evaluation practice systematically underweights the properties that make a model *interactive*. Broadly, there are two dominant evaluation families: *(i) perceptual fidelity under passive generation*, and *(ii) downstream task success under a policy*. Both are useful; neither isolates the fidelity of action-conditioned dynamics.

3.1. What benchmarks measure today

For video and image generation, benchmarks and metrics such as LPIPS, FVD, and VBench largely reward perceptual realism, temporal smoothness, and semantic alignment under prompt-only rollouts (Zhang et al., 2018; Unterthiner et al., 2019; Du et al., 2023). Recent long-horizon generation work explicitly targets extended coherence (Lu et al., 2024; Tan et al., 2024), reinforcing the idea that longer, smoother video implies a stronger world model. For embodied settings, benchmarks such as D4RL, Meta-World, BEHAVIOR, RLBench, ALFRED, Habitat, and Ego4D focus on end-task performance, instruction following, or embodied perception (Fu et al., 2021; Yu et al., 2020; Li et al., 2022; James et al., 2020; Shridhar et al., 2020; Savva et al., 2019; Grauman et al., 2022). These evaluations conflate multiple factors: the policy, exploration and data collection, reward structure, and environment design. Even when a learned world model is present, the benchmark signal is not diagnostic of whether the model's dynamics are correct.

Agent benchmarks for tool use and web interaction further expand the scope but also amplify the confound: results reflect the full stack (planning, tool use, prompting, web priors), not the causal fidelity of the underlying dynamics (Liu et al., 2023; Xu et al., 2023; Zhou et al., 2023). Recent work has started to formalize diagnostics for what it means for a generative model to "contain" an implicit world model (Vafa et al., 2024), but the community still lacks a standard substrate where action-conditioned transitions are explicitly specified and thus directly testable.

3.2. Action–state consistency: the interactive criterion

Action–state consistency asks two simple questions that are fundamental for interactive use: *(i) can we infer the action from the transition*, and *(ii) do different actions reliably induce different futures*? These questions capture causal legibility and controllable branching, two requirements for planning and verification. They are not equivalent to perceptual realism: a transition can look plausible yet correspond to no coherent intervention (action ambiguity), or distinct actions can lead to visually similar outcomes (branch collapse).

We emphasize that this is not merely an academic preference. In planning, an agent must reason about counterfactuals: "if I take action a instead of a', do I end up somewhere meaningfully different?" In safety and verification, we need to check whether interventions have predictable effects, and whether the model respects constraints (e.g., object permanence, conservation, contact). A model that fails these tests may still produce excellent videos, but it cannot serve as a reliable simulator for decision making.

3.3. A metric quad for interface-aware evaluation

To operationalize this criterion, we define a compact metric quad (Table 2) spanning the state and action axes, and the consistency vs. degrees-of-freedom axes. The quad is

Table 1. Taxonomy of world models across state modality and action interface. SC=State Consistency, SD=State DoF, AR=Action Rationality (AR_{inv}), AF=Action DoF. Scores: H=High, M=Medium, L=Low, F=Finite, V=Variable, –=N/A, 0=Zero.

Model	State	Action	SC	SD	AR	AF	Model	State	Action	SC	SD	AR	AF
Model-Based RL							***Interactive World Models***						
PlaNet (Hafner et al., 2019)	Latent	Motor	H	M	H	M	Genie 1 (Bruce et al., 2024a)	Video	Pixel	M	M	M	M
Dreamer (Hafner et al., 2020)	Latent	Motor	H	M	H	M	Genie 2/3	3D/Video	Pixel+NL	H	H	M	M
DreamerV2 (Hafner et al., 2021)	Latent	Symbolic	H	F	H	F	Oasis (Decart AI, 2024a)	Video	Pixel	M	M	M	M
DreamerV3 (Hafner et al., 2023)	Latent	Motor/Sym	H	V	H	V	GameNGen (Valevski et al., 2024a)	Video	Symbolic	H	F	H	F
MuZero (Schrittwieser et al., 2020)	Latent	Symbolic	H	V	H	F	Oh et al. (Oh et al., 2015)	Video	Symbolic	M	F	H	F
DIAMOND (Alonso et al., 2024)	Video	Symbolic	H	F	H	F							
Video Generation (Passive)							***Robotics & Physical AI***						
Sora (OpenAI, 2024a)	Video	None	H	H	L	0	RT-2 (Brohan et al., 2023a)	Image	Motor	V	M	H	M
Veo 2/3 (Google DeepMind, 2024)	Video	None	H	H	L	0	UniSim (Yang et al., 2024a)	Video	NL	H	M	H	M
VastDreamer (Wu et al., 2023)	3D/Video	None	H	H	L	0	Cosmos (NVIDIA Research, 2025b)	Video	Motor	H	H	H	H
FreeLong (Lu et al., 2024)	Video	None	H	H	L	0	OpenVLA (Kim et al., 2024)	Image	Motor	V	M	H	M
Video-Infinity (Tan et al., 2024)	Video	None	H	H	L	0	V-JEPA 2 (Bardes et al., 2025)	Video	Motor	H	M	H	M
Autonomous Driving							***LLM & Web-Based World Models***						
GAIA-1 (Hu et al., 2023a)	Video	Motor+Text	H	H	M	M	LLM World Sim (Wang et al., 2024)	Text	Symbolic	M	V	M	M
GAIA-2	Video	Motor+Text	H	H	M	M	Web World Models (Feng et al., 2025)	Text/Web	Sym+NL	H	V	H	M
							WebDreamer (Gu et al., 2025)	Text/HTML	Symbolic	M	M	H	M
3D World Generation							***History-Conditioned Video***						
World Labs (Li et al., 2024)	3D/Video	Continuous	H	F	H	F	PAN (Team, 2025b)	Video	NL	H	H	H	M
Representation Learning (State Encoders)							***Reference Environments***						
I-JEPA (Assran et al., 2023)	Latent	None	H	M	–	–	Classic Games (Mario)	2D Video	Symbolic	H	F	H	F
V-JEPA (Bardes et al., 2024)	Video	None	H	M	–	–	Minecraft	3D Block	Hybrid	M	∞	H	H
DINO (Caron et al., 2021)	Image	None	H	M	–	–	Real World	Physical	All	H	∞	H	∞
Proposed													
AC-World (ours)			Video + Graph			Hybrid				H	H	H	H

Table 2. Metric quad (overview).

	Consistency	Degree of Freedom
State ($\mathcal{S}$)	State Consistency (SC)	State DoF (SD)
Action ($\mathcal{A}$)	Action Rationality (AR_{inv})	Action DoF (AF)

intentionally interface-aware: it can be instantiated across modalities, but it must be evaluated within a fixed taxonomy cell.

State Consistency (SC)**.** Does the world hold together over time? SC captures temporal coherence and structural stability: objects do not teleport or morph without cause, camera pose does not drift arbitrarily, and long-horizon rollouts do not degrade into implausible artifacts. For visual modalities, SC can be instantiated by aggregating distributional and framewise measures (e.g., FVD, LPIPS-based stability, optical-flow consistency) (Unterthiner et al., 2019; Zhang et al., 2018). For text modalities, SC can be instantiated via contradiction/entailment checks and event-chain consistency.

State DoF (SD)**.** How rich is the reachable world? SD measures the diversity of valid reachable states under reasonable behavior. Operationally, one can embed reachable states (or rollouts) using fixed perceptual encoders such as CLIP or DINO (Radford et al., 2021; Caron et al., 2021), then cluster to estimate the number of distinct modes. This is not a perfect measure of openness, but it distinguishes collapsed worlds (few reachable modes) from worlds that support meaningfully different trajectories.

Action Rationality via Inverse Dynamics (AR_{inv})**.** Are transitions causally legible? AR_{inv} measures whether a transition is interpretable as the consequence of a specific intervention. Given a before/after pair (s_t, s_{t+1}) and a candidate action set $\mathcal{A}_{\text{cand}}$ (defined by the action interface), an inverse-dynamics judge Ψ predicts $\hat{a}_t$. We score

$$\mathsf{AR}_{\text{inv}} = \mathbb{E}_\tau[\mathbf{1}\{\hat{a}_t = a_t\}],$$

where low values indicate action ambiguity or inconsistent dynamics. Importantly, AR_{inv} is *ill-posed* when no candidate set exists (prompt-only generation), because there is no stable action vocabulary against which to evaluate invertibility. This is a central reason we argue that prompt-only rollouts cannot serve as a benchmark for interactive world modeling.

Action DoF (AF)**.** How many distinct futures can actions reliably induce from the same state? AF measures controllable branching: from the same state, do different actions induce meaningfully different outcomes? Operationally, from a fixed state s_t, sample a set of distinct candidate actions $\{a_t^{(n)}\}_{n=1}^N$, predict next states $\{\tilde{s}_{t+1}^{(n)}\}$, embed with a fixed encoder ϕ, and cluster outcomes to estimate the number of distinct modes. Low AF corresponds to "rail-shooter" dynamics where many actions collapse to the same future; high AF corresponds to sandbox-like control.

3.4. Why prompts are not actions

A natural objection is that "a prompt *is* an action." In an informal sense, it is: prompts influence the distribution of generated outputs. But as an *action interface* for benchmarking and control, prompts have three fundamental problems.

First, prompts do not define a closed, stable action set. Even if we restrict prompts to a template, small paraphrases can change semantics, and the space is unbounded. This makes inverse-dynamics evaluation ambiguous: without a candidate set, there is no well-posed notion of whether a transition is uniquely attributable to an intervention.

Second, prompts do not enforce stepwise compositionality. An interactive interface should support repeated application of actions with consistent semantics across states. Prompt-only generation typically treats each request as a new sample from a conditional distribution, without an explicit state that is updated by a fixed transition semantics. As a result, multi-step "rollouts" are often a sequence of loosely related generations rather than a coherent trajectory.

Third, prompt semantics are entangled with rendering. If the only way to induce a change is to rewrite a full scene description, then "action" becomes indistinguishable from "state specification." This collapses the distinction between dynamics and state representation, and makes it difficult to attribute failure: did the model misunderstand the intervention, or did it simply generate a different scene?

These issues motivate a practical stance: prompts can be *one* way to specify actions, but only when embedded in a structured interface that defines candidate actions, state updates, and evaluation protocols.

4. From Principle to Practice: An Action-Centric Construction

Our goal is not to claim a state-of-the-art model, but to show one concrete path that makes action–state consistency operational. We instantiate this path as explicit action-conditioned world graphs, then ground them into text, images, and videos.

4.1. Design Requirements

The thesis implies a minimal set of design requirements. (i) **An explicit action interface**: actions must be enumerable (or at least discretizable) so that evaluation can define candidate sets and compare outcomes. (ii) **Explicit, human-interpretable states**: states must be describable in a way that supports grounding into the chosen modality (text/image/video). (iii) **Graph-structured transitions**: branching should be an explicit object, not an emergent property of sampling. (iv) **Multi-ending outcomes**: many real tasks admit graded success and multiple failure modes; explicit endings enable richer evaluation than binary success/failure.

These requirements are intentionally weaker than "build a perfect simulator." They are sufficient to expose the failure modes we care about and to make action-centric diagnostics measurable.

4.2. Action-Conditioned World Representation

Referring to the definitions introduced in Section 1.1, we represent worlds as explicit action-conditioned graphs, following the AC-World formalism. Conceptually, an AC-World instantiates an MDP-like structure, but with one design choice that is crucial for diagnosis: each state is an *explicit, human-interpretable description* rather than an opaque symbol. A world is a controlled dynamical system

$$(\mathcal{S}, \mathcal{A}, \mathcal{T}, s_0, G, \mathcal{F}),$$

where we make the support of the transition mechanism explicit as a directed multigraph $\mathcal{T} \subseteq \mathcal{S} \times \mathcal{A} \times \mathcal{S}$. A trajectory is a finite path

$$\tau = \big((s_0, a_0, s_1), \ldots, (s_{T-1}, a_{T-1}, s_T)\big),$$

with $(s_t, a_t, s_{t+1}) \in \mathcal{T}$ and $s_T \in \mathcal{F}$. States are textual descriptions with structured metadata (progress, quality, tools), and actions are human-readable descriptions with optional type and risk labels. This makes both dynamics errors and policy errors diagnosable at the level of graph edges and branch choices.

States as anchors for grounding. Each state includes (i) a natural language description that can be used directly for text-only evaluation and (ii) structured metadata for task-relevant variables. In practical worlds (e.g., assembly, cooking, driving), metadata can encode progress, quality, constraints, and safety-critical indicators. These fields do not replace perception; they provide a scaffold that makes grounding and evaluation less ambiguous. For instance, a driving world can explicitly represent whether a seatbelt is buckled or whether the parking brake is engaged, enabling unambiguous action effects even when rendered visually.

Actions as causal abstractions. Actions are described at a human-interpretable granularity: they are intended to be inferable from state changes and stable across contexts. This differs from prompt-only control, where actions are often entangled with full-scene specification. It also differs from raw motor control, where action semantics may be too low-level for interpretability. In this sense, AC-World's actions are closer to the abstraction level of symbolic planning while remaining compatible with grounding into continuous visual change.

Graph diagnostics. Once transitions are explicit, the graph itself becomes analyzable. We can measure decision points (states with multiple outgoing actions), branching factor, path diversity, and the structure of failure basins. This enables a new kind of evaluation: we can distinguish whether a model fails because it cannot represent branching, because it violates constraints, or because it collapses multiple actions into a single future.

4.3. World Families and Decision Structure

We construct three families of worlds that probe increasing decision complexity while remaining interpretable. *Linear worlds* contain a single canonical path and primarily test temporal consistency and stepwise execution. *Branching worlds* introduce decision points where multiple admissible actions are available, testing whether a model can represent alternative futures and whether actions induce distinct outcomes. *Multi-ending worlds* enrich the graph with multiple terminal states of varying quality, including explicit failures, reflecting that many tasks have graded success and multiple ways to go wrong.

These families align with a practical view of benchmark design: difficulty is not only about long horizon, but also about the density and irreversibility of decisions. A world with a few high-stakes branch points can be more challenging (and more diagnostic) than a long linear procedure, because it requires causal reasoning and counterfactual prediction rather than memorization.

We emphasize that the graph formalism is compatible with both hand-curated worlds and LLM-generated worlds. LLM generation is useful for scaling diversity, while explicit graph validation prevents common failure modes such as unreachable states or inconsistent transitions.

4.4. Grounding into Textual, Egocentric, and Visual Modalities

We ground the same underlying action graph into multiple modalities, preserving transition structure across representations.

Text worlds. Text worlds can be generated by an LLM as a sequence of concrete state descriptions and actions, then expanded into branching and multi-ending variants by inserting alternative actions (shortcuts, risky choices, common mistakes) at selected states. Text worlds support fast iteration and allow evaluation of language-only world models or planners. They also provide a clean setting for action-inference diagnostics, because before/after states are explicit.

Egocentric worlds. For tasks where visual grounding matters, we rewrite state descriptions into strict egocentric, first-person keyframes: camera height and angle, visible hands and tools, object layout, and salient cues. Actions specify hand trajectories and object motion, reducing ambiguity and making visual grounding more consistent. This aligns with egocentric datasets and embodied perception benchmarks (Grauman et al., 2022; Goyal et al., 2017).

Image worlds via reference-based variation. Image worlds are generated by first rendering an initial state from scratch, then generating subsequent states via action-conditioned image variation using the previous image as a reference. This design is pragmatic: it enforces identity persistence (same objects, same layout, same camera) while isolating the visible change caused by the action. It directly supports evaluation of AR_{inv} (action inference from image pairs) and SC (temporal stability across transitions).

Video worlds via first/last-frame interpolation. Given images for states, transition videos are generated using first-frame + last-frame interpolation guided by an action prompt. This ensures that the video begins and ends at known states, making evaluation and debugging tractable. Importantly, grounding into video does not change the underlying graph: the world remains a set of discrete states with explicit transitions; the video is a rendering of an edge.

Implementation note: canonical vs. full-world grounding. In branching graphs, full grounding of all states and all transitions can be expensive. We therefore distinguish *canonical path* grounding (a main success path) from *full-world* grounding (BFS over the reachable graph). This separation makes it feasible to scale curation without paying the full cost of rendering every branch upfront, while still supporting on-demand diagnostics for branches that matter.

4.5. Protocols enabled by explicit graphs

Once the world is explicit, evaluation can separate the fidelity of a world model from the competence of a policy. We can evaluate in a teacher-forced setting (predict s_{t+1} given (s_t, a_t)) to test transition modeling, in an inverse setting (infer a_t from (s_t, s_{t+1})) to test causal legibility, and in a branching setting (compare outcomes under alternative actions) to test controllability. These protocols are difficult to define when the world is implicit in a generator and actions are not enumerable.

5. Evidence: What Breaks Without Action-Centricity

We include a single minimal diagnostic experiment that is feasible with our current implementation and that directly targets the action-centric failure mode: *action ambiguity*. The diagnostic is intentionally small and interpretable; the

Algorithm 1 Full-world image grounding via BFS over an action-conditioned graph.

```
Generate initial image for s_0 from scratch.
Initialize queue with outgoing transitions (s_0, a, s').
while queue not empty do
    Pop (s, a, s') from queue.
    if s' not yet rendered then
        Render s' by varying the image of s conditioned on a.
    end if
    Record the transition edge (s, a, s').
    Push all outgoing transitions from s' to queue.
end while
```

goal is to demonstrate that action-centric evaluation is well-posed and measurable, not to establish a leaderboard.

Minimal diagnostic: action inference on a small branching world. We use a small branching world (apple-eating) with 8 states and 7 transitions. Each state is a concrete, human-interpretable description of a tabletop scene; branches correspond to distinct interventions (e.g., *cut* vs. *bite*), and terminal states represent different outcomes. The task is: for each transition (s_t, a_t, s_{t+1}), infer a_t from the pair (s_t, s_{t+1}).

Protocol. For each transition, we define a candidate set $\mathcal{A}_{\text{cand}}$ consisting of (i) the outgoing actions available from s_t in the graph and (ii) a small set of hard negatives sampled from other parts of the world (actions that are plausible in the domain but not valid from s_t). We then ask an inverse-dynamics judge Ψ (an LLM for text states, or a VLM for image states) to select the most likely action. We report top-1 accuracy, which instantiates AR_{inv}.

Why this matters. If a world model collapses distinct interventions into visually similar outcomes, or produces transitions that could plausibly be explained by multiple actions, then AR_{inv} will be low. Conversely, when actions produce distinct and consistent effects, inverse inference becomes reliable. Prompt-only rollouts typically do not provide a stable $\mathcal{A}_{\text{cand}}$ because there is no fixed action vocabulary; as a result, the diagnostic becomes ill-posed or degenerates into free-form prompt matching.

Interpretation. This diagnostic does not require long-horizon rollouts; it is a one-step test of causal legibility. Its purpose is to surface failure modes that are invisible to perceptual metrics: a video can be smooth and realistic while still being action-ambiguous.

Table 3. Minimal action-inference diagnostic aligned to the current AC-World implementation.

World:	Apple-eating (branching)
States:	8
Transitions:	7
Metric:	Top-1 action-inference accuracy (AR_{inv})

6. Implications for the Field

Action-centric evaluation has immediate consequences for benchmarking, model design, and how the field communicates progress.

Benchmarks should separate dynamics from policies. Many benchmarks conflate environment dynamics with agent competence. An action-centric substrate enables teacher-forced evaluation of transition fidelity and inverse-dynamics evaluation of causal legibility, decoupling model errors from policy errors. This is especially important for multimodal agents where prompt engineering can mask dynamics failures.

World graphs as reusable testbeds. Explicit action graphs provide reusable, shareable environments that can be grounded into different modalities without changing underlying dynamics. This makes results comparable across model classes: a text-only planner, an image-conditioned model, and a video world model can be evaluated on the same graph, at the interface appropriate to each.

Co-evolution of models and policies becomes diagnosable. If world models and policies are trained together, it becomes hard to tell whether progress comes from better dynamics or better exploitation of quirks. Graphs make failure analysis concrete: did the policy choose a bad branch, or did the model mispredict an edge? This distinction matters for safety and for scientific understanding.

Representation learning is necessary but insufficient. Self-supervised representation learning and JEPA-style prediction (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024) provide powerful state abstractions and may improve stability and long-horizon prediction. But without explicit action interfaces and measurable branching, progress can still be mis-scored by passive realism. In other words: better latents help, but action grounding is the missing axis for benchmarking interactive correctness.

A call for standardization. We propose that future benchmark releases report results by taxonomy cell, include explicit candidate action sets, and publish graph-level statistics (branching factor, decision-point density, ending diversity). This would move the field from impressionistic comparisons to interface-aware, diagnosable evaluation.

7. Alternative Views and Counterarguments

"Prompts are actions." Prompts influence outputs, but as a benchmark interface they lack a stable, enumerable action set and do not enforce stepwise compositionality. Without a candidate set, inverse-dynamics evaluation is ill-posed; without step semantics, rollouts do not define a coherent transition system. Prompts can be part of an action interface (e.g., as a natural language action token), but only when embedded in a structured state/action protocol.

"Agents can learn around it." An agent may compensate for dynamics errors by exploiting biases or memorizing trajectories, especially in small benchmarks. This is precisely why evaluation should separate transition fidelity from policy success: otherwise we reward brittle strategies that do not generalize, and we fail to measure whether the world model itself is correct.

"Realism correlates with usefulness." Realism may correlate with some forms of usefulness, but correlation is not evaluation. A benchmark must support falsifiable diagnostics: it must tell us when action-conditioned dynamics are wrong, not merely when outputs look plausible. Action-centric metrics complement realism metrics by measuring what realism does not: causal legibility and controllable branching.

"Physical consistency should be the priority." Physical consistency is necessary, but many physically plausible futures are compatible with the same perceptual stream unless actions are specified. Action–state consistency does not replace physical evaluation; it makes it conditional on interventions, which is the regime where agents operate.

8. Conclusion

We argued that progress in world models should be measured not by how real the world looks, but by how reliably it responds to intervention. A state-modality $\times$ action-interface taxonomy makes comparisons well-posed, a compact metric quad exposes the missing evaluation axis, and an explicit action-graph construction makes the principle operational across modalities. Our minimal diagnostic illustrates that action-centric evaluation is feasible today and surfaces failure modes invisible to passive perceptual metrics. If the field wants world models that support planning, tool use, and verification, it should benchmark what agents need: action–state consistency.

References

Alonso, E., Jelley, A., Micheli, V., Kanervisto, A., Storkey, A., Pearce, T., and Fleuret, F. Diffusion for world modeling: Visual details matter in atari. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL https://arxiv.org/abs/2405.12399. Spotlight.

Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., and Ballas, N. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. URL https://openaccess.thecvf.com/content/CVPR2023/papers/Assran_Self-Supervised_Learning_From_Images_With_a_Joint-Embedding_Predictive_Architecture_CVPR_2023_paper.pdf.

Bardes, A., Garrido, Q., Balestriero, R., Chen, J., Bardes, a., et al. Revisiting feature prediction for learning visual representations from video. *arXiv preprint arXiv:2404.08471*, 2024. URL https://arxiv.org/abs/2404.08471.

Bardes, A. et al. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint*, 2025. URL https://ai.meta.com/research/publications/v-jepa-2/.

Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Chen, X., Choromanski, K., et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023a. URL https://arxiv.org/abs/2307.15818.

Brohan, A. et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control, 2023b. URL https://arxiv.org/abs/2307.15818. Alias for brohan2023rt2.

Bruce, J., Dennis, M., Edwards, A., Parker-Holder, J., Shi, Y., Hughes, E., Lai, M., Mber, A., Dasgupta, R., Lohman, R., Bengio, E., Clark, C., de Freitas, N., et al. Genie: Generative interactive environments. *arXiv preprint arXiv:2402.15391*, 2024a. URL https://arxiv.org/abs/2402.15391.

Bruce, J. et al. Genie: Generative interactive environments, 2024b. URL https://arxiv.org/abs/2402.15391. Alias for bruce2024genie.

Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., and Joulin, A. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. URL https://openaccess.thecvf.com/content/ICCV2021/papers/Caron_Emerging_Properties_in_Self-Supervised_Vision_Transformers_ICCV_2021_paper.pdf.

Decart AI. Oasis: A universe in a transformer. *Decart Technical Report*, 2024a. URL https://decart.ai/publications/oasis-interactive-ai-video-game-model.

Decart AI. Oasis: A universe in a transformer, 2024b. URL https://decart.ai/publications/oasis-interactive-ai-video-game-model. Alias for decart2024oasis.

Du, Y. et al. Vbench: Comprehensive benchmark suite for video generative models. *arXiv*, 2023.

Feng, J., Zhang, Y., Zhang, C., Lu, Y., Liu, S., and Wang, M. Web world models. *arXiv preprint arXiv:2512.23676*, 2025. URL https://arxiv.org/abs/2512.23676.

Fu, J., Kumar, A., Nachum, O., Tucker, G., and Levine, S. D4rl: Datasets for deep data-driven reinforcement learning. *NeurIPS*, 2021.

Google DeepMind. Veo: State-of-the-art video generation, 2024. URL https://deepmind.google/models/veo/.

Goyal, R. et al. The "something something" video dataset for learning and evaluating visual common sense. *ICCV*, 2017.

Grauman, K. et al. Ego4d: Human-centric video understanding at scale. *CVPR*, 2022.

Gu, Y., Zheng, B., et al. Is your llm secretly a world model of the internet? model-based planning for web agents. *Transactions on Machine Learning Research (TMLR)*, 2025. URL https://arxiv.org/abs/2411.06559.

Hafner, D., Lillicrap, T., Norouzi, M., and Ba, J. Learning latent dynamics for planning from pixels. *ICML*, 2019.

Hafner, D., Lillicrap, T. P., Ba, J., and Norouzi, M. Dream to control: Learning behaviors by latent imagination. *ICLR*, 2020.

Hafner, D., Pasukonis, J., Ba, J., and Lillicrap, T. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023. URL https://arxiv.org/abs/2301.04104.

Hafner, D. et al. Mastering atari with discrete world models. *ICLR*, 2021.

Hu, A., Russell, L., Yeo, H., Murez, Z., Fedoseev, G., Kendall, A., Hafner, J., and Sherrod, V. Gaia-1: A generative world model for autonomous driving. *arXiv preprint arXiv:2309.17080*, 2023a. URL https://arxiv.org/abs/2309.17080.

Hu, A. et al. Gaia-1: A generative world model for autonomous driving, 2023b. URL https://arxiv.org/abs/2309.17080. Alias for hu2023gaia1.

James, S., Ma, Z., and Rosman, B. Rlbench: The robot learning benchmark & learning environment. In *IEEE RA-L*, 2020.

Ke, T.-H. et al. A survey of world models. *arXiv*, 2023.

Kim, M. J., Pertsch, K., Karamcheti, S., et al. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024. URL https://arxiv.org/abs/2406.09246.

LeCun, Y. A path towards autonomous machine intelligence (version 0.9). 2022. URL https://openreview.net/pdf?id=BZ5a1r-kVsf.

Li, C. et al. Behavior: Benchmark for everyday household activities in virtual, interactive, and ecological environments. In *CVPR*, 2022.

Li, F.-F., Johnson, J., Lassner, C., and Mildenhall, B. World labs: Spatial intelligence for 3d world generation, 2024. URL https://www.worldlabs.ai/.

Liu, X. et al. Agentbench: Evaluating llms as general-purpose agents. *arXiv*, 2023.

Lu, Y., Liang, Y., Zhu, L., and Yang, Y. Freelong: Training-free long video generation with spectralblend temporal attention. *arXiv preprint arXiv:2407.19918*, 2024. URL https://arxiv.org/abs/2407.19918.

NVIDIA Research. Cosmos world foundation model platform for physical ai, 2025a. URL https://research.nvidia.com/publication/2025-01_cosmos-world-foundation-model-platform-physical-a Alias for nvidia2025cosmos.

NVIDIA Research. Cosmos world foundation model platform for physical ai. *NVIDIA Technical Report*, 2025b. URL https://research.nvidia.com/publication/2025-01_cosmos-world-foundation-model-platform-physical-a

Oh, J., Guo, X., Lee, H., Lewis, R. L., and Singh, S. Action-conditional video prediction using deep networks in atari games. In *NeurIPS*, 2015.

OpenAI. Video generation models as world simulators. *OpenAI Technical Report*, 2024a. URL https://openai.com/index/video-generation-models-as-world-simulators/.

OpenAI. Video generation models as world simulators, 2024b. URL https://openai.com/index/video-generation-models-as-world-simulators/. Alias for openai2024sora.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., and Sutskever, I. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2021.

Savva, M. et al. Habitat: A platform for embodied ai research. In *ICCV*, 2019.

Schrittwieser, J., Antonoglou, I., Hubert, T., Simonyan, K., Sifre, L., Schmitt, S., Guez, A., Lockhart, E., Hassabis, D., Graepel, T., Lillicrap, T., and Silver, D. Mastering atari, go, chess and shogi by planning with a learned model. *Nature*, 588(7839):604–609, 2020. URL https://arxiv.org/abs/1911.08265.

Shridhar, M. et al. Alfred: A benchmark for interpreting grounded instructions. In *CVPR*, 2020.

Tan, Z., Yang, X., Liu, S., and Wang, X. Video-infinity: Distributed long video generation. *arXiv preprint arXiv:2406.16260*, 2024. URL https://arxiv.org/abs/2406.16260.

Team, M. I. Pan: A world model for general, interactable, and long-horizon world simulation, 2025a. URL https://arxiv.org/abs/2511.09057. Alias for pan2025worldmodel.

Team, M. I. Pan: A world model for general, interactable, and long-horizon world simulation. *arXiv preprint arXiv:2511.09057*, 2025b. URL https://arxiv.org/abs/2511.09057.

Unterthiner, T., Kornblith, S., Kushman, N., Zaremba, W. J., Bojarski, A. M. P., and et al., A. D. Towards accurate generative models of video: A new metric and challenges. arXiv preprint arXiv:1812.01717, 2019.

Vafa, K., Chen, J. Y., Rambachan, A., Kleinberg, J., and Mullainathan, S. Evaluating the world model implicit in a generative model. *arXiv preprint arXiv:2406.03689*, 2024. URL https://arxiv.org/abs/2406.03689.

Valevski, D., Leviathan, Y., Arar, M., and Fruchter, S. Diffusion models are real-time game engines. *arXiv preprint arXiv:2408.14837*, 2024a. URL https://arxiv.org/abs/2408.14837.

Valevski, D. et al. Diffusion models are real-time game engines, 2024b. URL https://arxiv.org/abs/2408.14837. Alias for valevski2024gamengen.

Wang, R., Todd, G., Yuan, E., Xiao, Z., Gartner, R., Cote, M.-A., Clark, P., and Clark, C. Can language models serve as text-based world simulators? In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. URL https://arxiv.org/abs/2406.06485.

Wu, Y. et al. Vastdreamer: Universal world models for 3d video generation. *arXiv*, 2023.

Xu, G. et al. Toolbench: A large-scale dataset for training and evaluating tool-using agents. *arXiv*, 2023.

Yang, S., Du, Y., Ghasemipour, K., Tompson, J., Kaelbling, L., Schuurmans, D., and Abbeel, P. Learning interactive real-world simulators. In *International Conference on Learning Representations (ICLR)*, 2024a. URL https://arxiv.org/abs/2310.06114. Outstanding Paper Award.

Yang, S. et al. Learning interactive real-world simulators, 2024b. URL https://arxiv.org/abs/2310.06114. Alias for yang2024unisim.

Yu, T. et al. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *CoRL*, 2020.

Zhang, R., Isola, P., Efros, A. A., Shechtman, E., and Wang, O. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

Zhou, S. et al. Webarena: A realistic web environment for building autonomous agents. *arXiv*, 2023.